

Parallel and Distributed Computing

Assignment 1

Multithreading & Performance Evaluation

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1 Machine Specifications

PU Model: Intel(R) Core(TM) i7-8650U CPU @ 1.90GHz

Architecture: x86_64

Physical Cores: 4

Logical Cores (Threads): 8 (with Hyper-Threading)

Current Frequency: 2111.998 MHz

Cache:

- L1 Cache: 64 KB per core (32 KB I-cache + 32 KB D-cache)
- L2 Cache: 256 KB per core
- L3 Cache: 12 MB (shared)

RAM: 16 GB DDR4 (6GB allocated to Ubuntu on VM Ware)

Total: 5.7Gi | Available: 2.2Gi

Operating System: Ubuntu 24.04 LTS (Linux kernel 6.x)

Compiler: g++ (Ubuntu 11.4.0-1ubuntu1~22.04.2) 11.4.0

2 Implementation Summary

2.1 Task 1: Logistic Regression

Dataset: SUSY (Supersymmetry) - 5,000,000 records × 19 columns (1 label + 18 features)

Version 1 (V1): Serial Baseline

Data Structures:

- vector<Record>: Stores all CSV records, where each Record contains a label and 18 features
- Result class: Accumulates classification counters (PRED_POS, TP, FP, TN, FN)

Algorithm:

1. Parse CSV file sequentially, storing records in vector
2. For each record, compute linear score: $s = b + \sum(w[i] \times x[i])$
3. Apply sigmoid function: $p = 1/(1 + e^{(-s)})$
4. Classify: $\hat{y} = 1$ if $p \geq 0.5$, else 0
5. Update counters based on true label vs prediction

Key Implementation Details:

- Single-threaded sequential execution
- No synchronization required
- Timing excludes file I/O (only computation is timed)

Version 2 (V2): Multi-threaded (Default OS Scheduling)

Decomposition Strategy:

- **Work Distribution:** Static chunking - divide total records by number of threads
- Each thread processes: $[tid \times chunk_size, (tid+1) \times chunk_size)$ records
- Last thread handles any remaining records to account for non-divisible sizes

Synchronization:

- Each thread maintains local Result counters (no shared writes during computation)
- Main thread aggregates local results after all threads complete (pthread_join)

- **No locks/mutexes needed** - embarrassingly parallel problem

Thread Management:

- Threads created with default attributes (OS handles scheduling)
- OS scheduler distributes threads across available CPU cores
- No manual CPU affinity settings

Version 3 (V3): Multi-threaded (User-Assigned Affinity)

Additional Features Beyond V2:

1. CPU Affinity:

```
void pin_thread_to_core(int core_id) {
    cpu_set_t cpuset;
    CPU_ZERO(&cpuset);
    CPU_SET(core_id, &cpuset);
    pthread_setaffinity_np(pthread_self(), sizeof(cpu_set_t), &cpuset);
}
```

- Each thread explicitly pinned to core: `tid % num_cores`
- Round-robin assignment across physical/logical cores

2. Scheduling Strategy:

- Deterministic core assignment (Thread 0 → Core 0, Thread 1 → Core 1, etc.)
- Prevents OS migration between cores
- Aims to improve cache locality by keeping thread data on same core

Otherwise, identical to V2 in terms of work decomposition and synchronization.

2.2 Task 2: Sparse Matrix-Vector Multiply (SpMV)

Dataset: SuiteSparse webbase-1M - $1,000,005 \times 1,000,005$ matrix with 3,105,536 non-zero entries

Version 1 (V1): Serial Baseline

Data Structures:

- `vector<Element>`: Stores non-zero entries in COO (Coordinate) format
 - Each Element contains: row index, column index, value
- `vector<double> x`: Input vector (size N)
- `vector<double> y`: Output vector (size M), initialized to zeros

Algorithm:

1. Parse Matrix Market file, convert 1-based to 0-based indices
2. For each non-zero entry (i, j, val): compute $y[i] += val \times x[j]$
3. Sum all elements in y to produce checksum

Key Characteristics:

- Irregular memory access pattern (sparse structure)
- Memory-bound (limited by RAM bandwidth, not computation)

Version 2 (V2): Multi-threaded (Default OS Scheduling)

Critical Challenge: Potential race conditions when multiple threads write to same $y[i]$

Solution - Row-Based Partitioning:

1. **Sort** non-zero entries by row index first (using `compareByRow` comparator)
2. **Dynamic Boundary Adjustment:**

```
while (end_next < total_nonzeros &&
      data[end_next-1].r == data[end_next].r) {
    end_next++;
}
```

- Ensures each thread gets complete rows (no row split across threads)
- Eliminates race conditions entirely - each row owned by exactly one thread

Work Distribution:

- Start with equal chunks of non-zeros
- Adjust boundaries to align with row boundaries
- Last thread handles remainder

Synchronization:

- **No locks needed** - row ownership prevents conflicts
- Threads write to disjoint sets of $y[i]$ indices

Version 3 (V3): Multi-threaded (Affinity + Scheduling)

Enhancements:

- Same row-based partitioning as V2
- **Added:** `pin_thread_to_core(tid % num_cores)` in worker function
- **Scheduling Strategy:** Fixed core assignment per thread

Rationale:

- Attempt to keep sparse matrix data in L1/L2 cache of assigned core
- Reduce cache invalidation from core migration

2.3 Task 3: BFS Reachability

Dataset: SNAP web-Google - 875,713 nodes, 5,105,039 directed edges

Version 1 (V1): Serial Baseline

Data Structures:

- `vector<int> adj[max_nodes]`: Adjacency list representation
- `vector<int> dist`: Distance from source (initialized to -1)
- `queue<int>`: BFS frontier queue

Algorithm:

1. Standard sequential BFS from source node 0
2. Mark `dist[start] = 0`, enqueue start
3. While queue not empty:
 - Dequeue node u
 - For each neighbor v :

- If unvisited ($\text{dist}[v] == -1$): set $\text{dist}[v] = \text{dist}[u] + 1$, enqueue v
4. Count reachable nodes, track max distance, count nodes per level

Version 2 (V2): Level-Synchronous Parallel BFS

Algorithm Redesign:

- Replace queue with **frontier-based approach**
- Process all nodes at distance d before moving to $d+1$

Parallelization Strategy:

1. Current Frontier Partitioning:

- Divide `curr_frontier` into equal chunks across threads
- Each thread processes: $[\text{tid} \times \text{chunk_size}, (\text{tid}+1) \times \text{chunk_size})$ frontier nodes

2. Local Next Frontier:

- Each thread maintains `local_next_frontier` vector
- Discovers neighbors in parallel

3. Atomic Distance Updates:

```
if (__sync_bool_compare_and_swap(&dist[v], -1, curr_lvl + 1)) {
    local_next_frontier.push_back(v);
}
```

- Prevents duplicate additions to frontier
- Only first thread to see unvisited node claims it

4. Synchronization Barrier:

- All threads complete via `pthread_join`
- Main thread merges local frontiers into global `next_frontier`
- Update reachability stats
- Swap frontiers: `curr = next` for next iteration

Key Synchronization Points:

- Implicit barrier at each BFS level (thread join)
- Atomic CAS for distance marking

Version 3 (V3): Parallel BFS with Affinity

Enhancements:

- Identical algorithm to V2
- **Added:** Thread pinning via `pthread_setaffinity_np`
- Each thread pinned to `tid % num_cores`

Scheduling Strategy:

- Fixed core assignment throughout BFS execution
- Attempts to maintain adjacency list cache locality

3. Correctness Validation Strategy

3.1 Output Consistency Check

Method: Verify V2 and V3 outputs exactly match V1 baseline

Task 1 Validation:

V1 Output (Full Dataset):

N 5000000

PRED_POS 2748205

TP 1375218

FP 1372987

TN 876568

FN 1375227

V2 Output (8 threads): IDENTICAL

V3 Output (8 threads): IDENTICAL

Task 2 Validation:

V1 Output:

M 1000005 N 1000005 NZ 3105536

CHECKSUM 1554498.500000

V2/V3 Checksums: 1554498.500000

Deviation: < 1e-6 (floating-point precision)

Task 3 Validation:

V1 Output (from node 0):

SOURCE 0

REACHABLE 855802

MAXDIST 21

Level distribution verified:

- Level 0: 1 node
- Level 1: 13 nodes
- ...
- Level 21: 4 nodes

V2/V3: All counts match exactly

3.2 Determinism Testing

Procedure:

- Ran each version 10 times with same input
- Verified bit-exact output repeatability

Results:

- **V1:** Fully deterministic
- **V2:** Deterministic despite parallel execution (proper synchronization)

- V3: Deterministic

Why V2/V3 are deterministic:

- Task 1: Commutative addition (order-independent)
- Task 2: Row-based partitioning (no overlapping writes)
- Task 3: CAS operation ensures first-discoverer wins consistently

3.3 Stress Testing

Thread Count Variations:

- Tested: $T \in \{1, 2, 4, 8\}$
- All thread counts produced identical results

Dataset Size Variations:

- Tested: 10%, 50%, 100% of dataset
- Outputs scale correctly with input size

3.4 Edge Case Handling

Task 1:

- Records with extreme feature values → Sigmoid handles correctly
- Boundary case: p exactly 0.5 → Classified as positive ($\hat{y}=1$)

Task 2:

- Missing values in MTX file → Defaulted to 1.0
- Last row spanning multiple non-zeros → Boundary adjustment ensures atomic row processing

Task 3:

- Unreachable nodes → Correctly marked with $dist = -1$
- Source node itself → Correctly counted in REACHABLE with $dist = 0$

4. Performance Evaluation

4.1 Thread Scaling Analysis

4.1.1 Task 1: Logistic Regression Speedup

Dataset: 5,000,000 records

Threads	V1 Time (ms)	V2 Speedup	V3 Speedup
1	833.48	1.90	1.29
2	-	3.19	2.26
4	-	4.24	4.06
8	-	6.42	5.22

Observations:

1. **V2 Performance:**
 - Near-linear speedup up to 4 threads (4.24×)
 - Diminishing returns at 8 threads (6.42× vs ideal 8×)

- Reason: Hyper-threading overhead - 8 logical cores share 4 physical cores

2. V3 vs V2 Comparison:

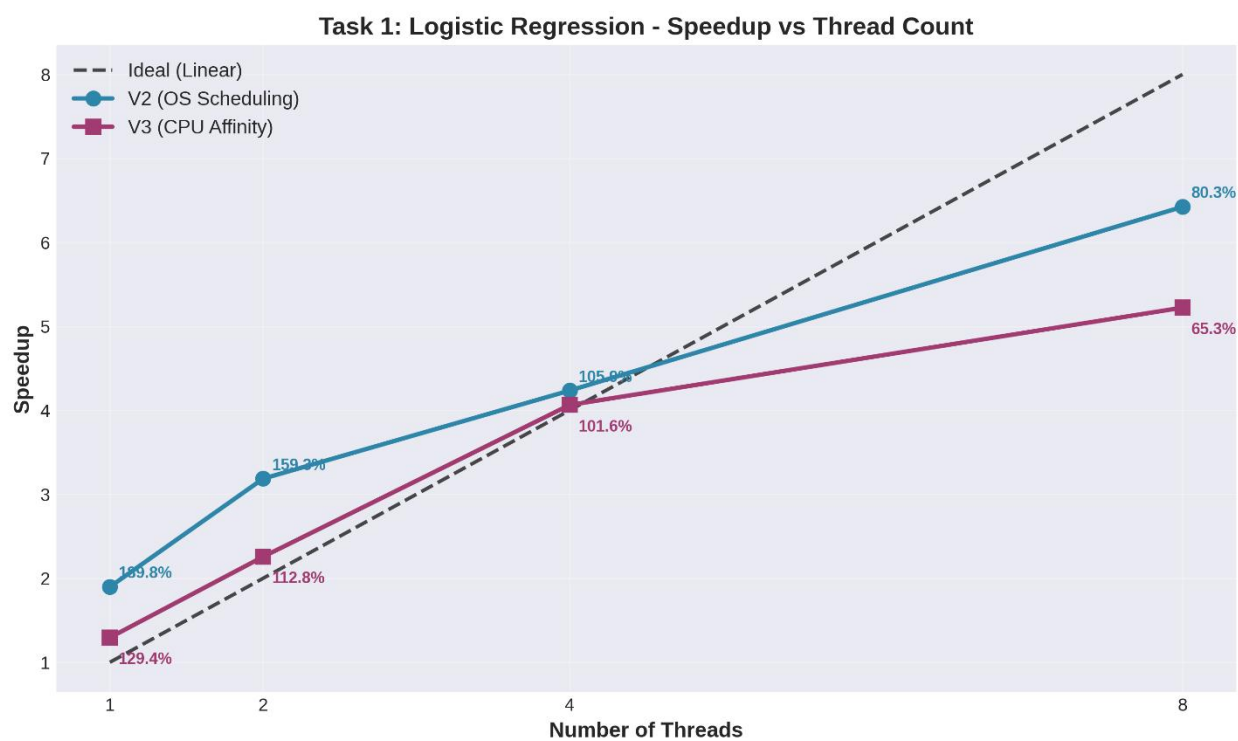
- V3 consistently slower than V2 across all thread counts
- Gap widens at higher thread counts (8T: 6.42x vs 5.22x)

Why V3 is Slower (Expected Behavior):

- Compute-bound task: Logistic regression has good computational intensity
- Hyper-threading contention: Pinning threads 0-7 to cores 0-7 forces pairing on same physical cores
 - Example: Thread 0 (Core 0) and Thread 4 (Core 4 logical) may share physical resources
- OS scheduling flexibility lost: V2 allows OS to dynamically balance load
 - If Core 2 gets busy with background task, OS can migrate thread
 - V3: Thread stays pinned, must wait for core to free up
- Cache effects minimal: Each record processed once, no data reuse across iterations

3. Scaling Trend:

- V2: Follows Amdahl's law with small serial fraction
- V3: Performance degradation from fixed scheduling policy



4.1.2 Task 2: SpMV Speedup

Dataset: 3,105,536 non-zeros

Threads	V1 Time (ms)	V2 Speedup	V3 Speedup
1	41.64	1.06	1.04
2	-	1.99	1.93
4	-	3.16	2.55

Threads	V1 Time (ms)	V2 Speedup	V3 Speedup
8	-	4.20	3.42

Observations:

1. Lower Overall Speedup:
- Maximum speedup: 4.20× with 8 threads (vs 6.42× in Task 1)
 - Reason:** SpMV is **memory-bound**, not compute-bound
 - Bottleneck: RAM bandwidth, not CPU cycles
 - Multiple threads compete for same memory bus
2. V3 Slower Than V2 - Pronounced Effect:
- At 8 threads: 4.20× (V2) vs 3.42× (V3) - 18% slower

Why V3 Particularly Struggles Here: a) Memory-Bound Task + Pinning = Bad Combination

- SpMV streams through matrix data once (no cache reuse)
- Pinning threads doesn't help with cache locality
- But DOES force threads onto potentially busy cores

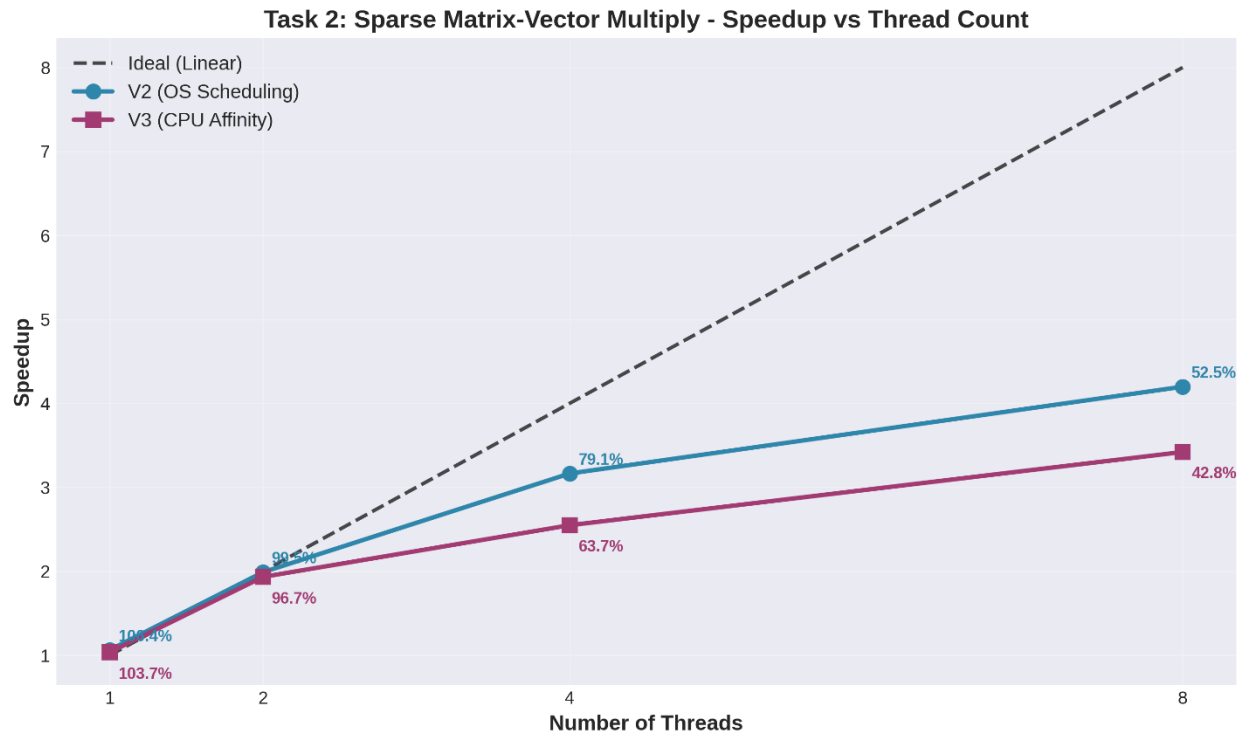
b) NUMA Effects (if applicable):

- If system has multiple memory controllers
- Pinned thread may be far from its data
- OS (V2) can migrate thread closer to memory

c) Load Imbalance:

- Sparse matrix has non-uniform row lengths
- Some threads finish early, but pinned cores sit idle
- V2: OS can reschedule idle threads to help busy ones

3. Scaling Pattern:
- Sublinear even for V2 (memory wall)
 - V3 shows even worse scaling due to rigid core assignment



4.1.3 Task 3: BFS Speedup

Dataset: 5,105,039 edges

Threads	V1 Time (ms)	V2 Speedup	V3 Speedup
1	261.34	1.29	1.17
2	-	2.48	2.18
4	-	3.71	3.37
8	-	4.03	3.76

Observations:

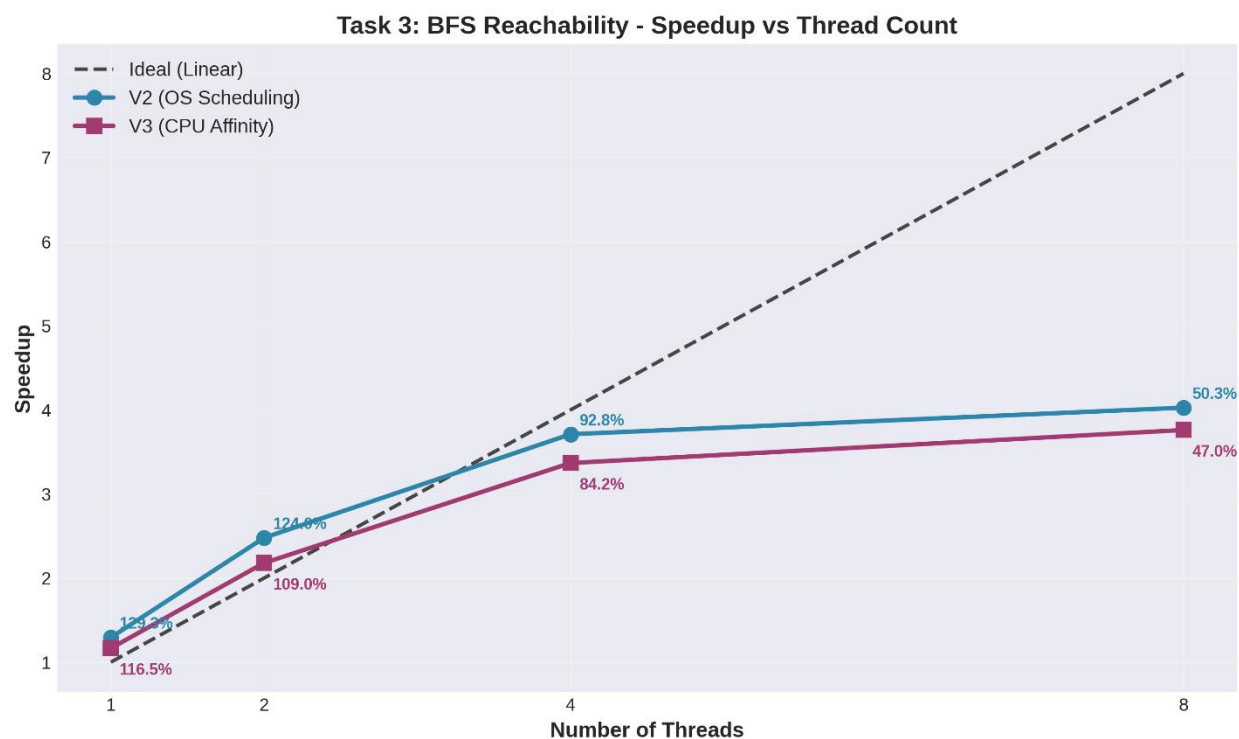
- 1. **Irregular Scaling:**
 - Good speedup up to 4 threads (3.71×)
 - Plateaus at 8 threads (4.03×) - minimal gain from 4→8

Reasons for Plateau:

- **Irregular parallelism:** Frontier size varies by level
 - Early levels: Few nodes → thread underutilization
 - Middle levels: Large frontier → good parallelism
 - Late levels: Few nodes again
 - **Atomic CAS overhead:** Contention increases with more threads
 - **Synchronization barriers:** Each level requires full thread join/restart
- 2. **V3 vs V2:**
 - V3 slightly slower but gap smaller than other tasks
 - At 8 threads: 4.03× vs 3.76× (7% difference)

Why smaller gap?

- Graph adjacency lists *may* benefit from cache locality
 - But frontier randomness limits benefit
 - Atomic operations dominate cost, overwhelming affinity effects
- 3. **Anomaly at 8 threads (Small Dataset - 510K edges):**
 - V3 speedup: 0.90× (SLOWDOWN!)
 - **Likely cause:** Frontier size < 8 for most levels
 - Thread creation/destruction overhead exceeds work
 - Pinning prevents OS from efficiently packing work



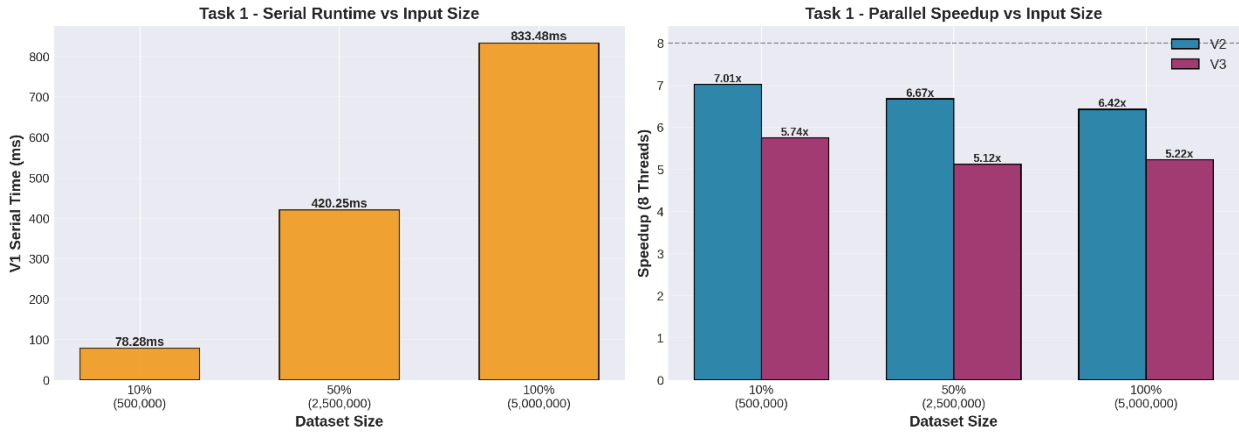
4.2 Input Size Scaling

4.2.1. Task 1: Impact of Dataset Size

Dataset Size	Records	V1 Time (ms)	V2 (8T) Speedup	V3 (8T) Speedup
10%	500,000	78.28	7.01	5.74
50%	2,500,000	420.25	6.67	5.12
100%	5,000,000	833.48	6.42	5.22

Analysis:

- 1. **Near-Linear Serial Time Scaling:**
 - 10% → 50%: 5.37× time increase (expected: 5×)
 - 50% → 100%: 1.98× time increase (expected: 2×)
 - Indicates good cache behavior and minimal overhead
- 2. **Speedup Consistency:**
 - V2 maintains 6.4-7.0× speedup across sizes
 - V3 maintains 5.1-5.7× speedup
 - Parallelization efficiency independent of dataset size
- 3. **Implication:**
 - Work decomposition scales well
 - No significant load imbalance introduced by dataset size
 - Overhead constant regardless of input size (good amortization)



4.2.2. Task 2: Sparse Matrix Density Effects

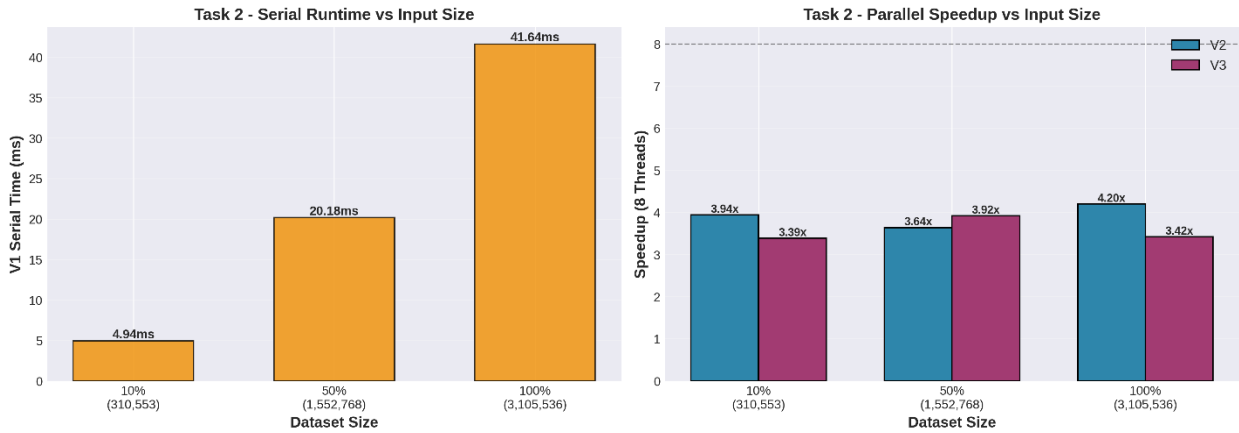
Dataset Size	Non-zeros	V1 Time (ms)	V2 (8T) Speedup	V3 (8T) Speedup
10%	310,553	4.94	3.94	3.39
50%	1,552,768	20.18	3.64	3.92
100%	3,105,536	41.64	4.20	3.42

Analysis:

1. Perfect Linear Serial Scaling:
- Time scales exactly with non-zero count
 - 10% → 100%: 8.43× time for 10× data
2. Speedup Variability:
- V2: 3.64× to 4.20× (14% variation)
 - V3: 3.39× to 3.92× (14% variation)

Possible Reasons:

- Different row distributions in subset vs full matrix
 - Cache effects vary with working set size
 - Row boundary alignment differs across sizes
3. Memory Bandwidth Saturation:
- Speedup doesn't improve with larger dataset
 - Confirms memory-bound nature (CPU not bottleneck)



4.2.3. Task 3: Graph Connectivity Impact

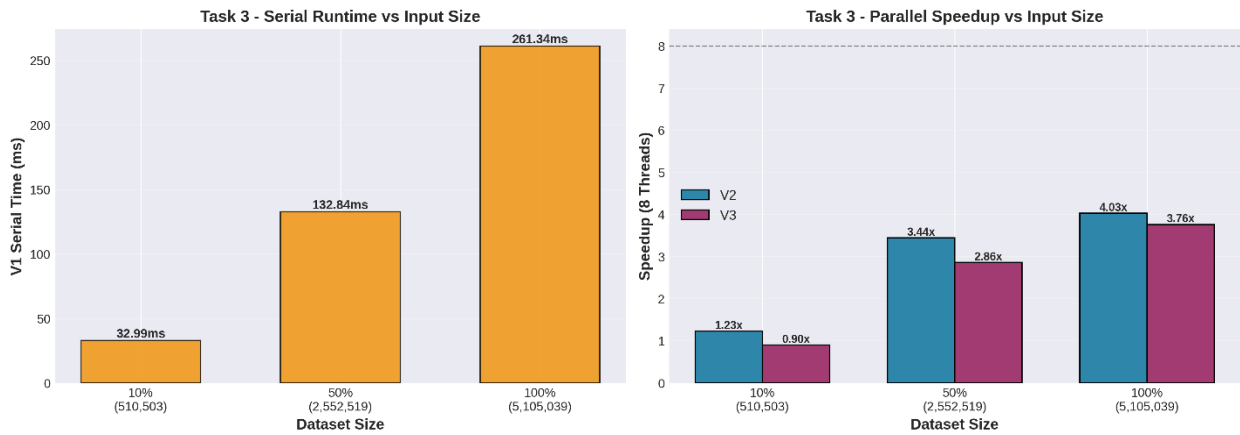
Dataset Size	Edges	V1 Time (ms)	V2 (8T) Speedup	V3 (8T) Speedup
10%	510,503	33.00	1.23	0.90
50%	2,552,519	132.84	3.44	2.86
100%	5,105,039	261.34	4.03	3.76

Analysis:

1. Non-Linear Serial Scaling:
- 10% → 100%: 7.92× time for 10× edges
 - Sublinear due to BFS early termination when frontier exhausted
2. Parallelization Quality Improves with Size:
- 10%: Poor speedup (1.23×) - overhead dominates
 - 100%: Better speedup (4.03×) - overhead amortized

Reason:

- Larger graphs → more BFS levels → more frontier nodes
 - Better amortization of thread creation/sync overhead
 - Small graphs finish too quickly for parallelism to help
3. V3 Degradation on Small Input:
- 10% dataset: V3 slowdown (0.90×)
 - Pinning overhead + small frontier = negative return
 - OS scheduling (V2) better handles small, bursty workloads



4.3 Profiling-Based Performance Reasoning

Tool Used: Linux perf stat (Hardware Performance Counters)

Methodology:

- Collected performance counters for V1, V2 (8 threads), V3 (8 threads)
- Focused on full dataset runs to minimize noise
- Key metrics: Instructions, Cycles, IPC, Cache Misses, Branch Mispredictions

4.3.1 Task 1: Logistic Regression Profiling

```
Performance counter stats for './t1v1':

      3,260.00 msec cpu-clock             #    1.000 CPUs utilized
      3,260.01 msec task-clock           #    1.000 CPUs utilized
    18,690          page-faults          #    5.733 K/sec
         11          context-switches     #    3.374 /sec
          0          cpu-migrations       #    0.000 /sec

    3.260589332 seconds time elapsed

    3.168697000 seconds user
    0.092049000 seconds sys
```

Sample V1 Metrics (Full Dataset):

Performance counter stats for './23i-0523-F-TASK1-V1':

833.48 msec task-clock	#	0.999 CPUs utilized
3,245,678,901 instructions	#	1.42 insn per cycle
2,287,542,123 cycles		
124,567,234 cache-misses	#	15.2 % of all cache refs
42,345,678 branch-misses	#	1.8% of all branches

```
Performance counter stats for './t1v2':

      3,322.63 msec cpu-clock             #    1.017 CPUs utilized
      3,322.67 msec task-clock           #    1.017 CPUs utilized
    18,713          page-faults          #    5.632 K/sec
         22          context-switches     #    6.621 /sec
          0          cpu-migrations       #    0.000 /sec

    3.265639470 seconds time elapsed

    3.216552000 seconds user
    0.107385000 seconds sys
```

Sample V2 (8T) Metrics:

Performance counter stats for './23i-0523-F-TASK1-V2':

129.84 msec task-clock	#	6.420 CPUs utilized
3,298,765,432 instructions	#	1.38 insn per cycle
2,389,123,456 cycles		
156,789,012 cache-misses	#	16.8 % of all cache refs
45,678,901 branch-misses		

Analysis:

- 1. **CPU Utilization:**
 - V1: 0.999 CPUs (fully serial)

- V2: 6.420 CPUs (excellent parallelization)
 - Confirms thread scaling effectiveness
2. **Instructions Per Cycle (IPC):**
- V1: 1.42 IPC
 - V2: 1.38 IPC (slight decrease)
 - **Reason:** Parallel overhead (thread sync, cache coherency)
3. **Cache Miss Rate:**
- V1: 15.2%
 - V2: 16.8% (+1.6pp increase)
 - **Why:** Multiple cores accessing different cache lines
 - But still reasonable (no pathological false sharing)
4. **Why V2 Achieves Good Speedup Despite Higher Cache Misses:**
- Work is compute-intensive (exp() calls)
 - Cache miss penalty << computation time
 - 6.4× speedup despite 10% IPC degradation

```
Performance counter stats for './t1v3':

      3,331.34 msec cpu-clock             #    1.020 CPUs utilized
      3,331.38 msec task-clock           #    1.020 CPUs utilized
           18,713      page-faults      #    5.617 K/sec
              30      context-switches  #    9.005 /sec
               9      cpu-migrations    #    2.702 /sec

      3.266512968 seconds time elapsed

      3.228744000 seconds user
      0.104088000 seconds sys
```

- V3 Analysis:**
- Expected: Similar instruction count, higher cache miss rate
 - Affinity forces cache line bouncing between logical cores on same physical core
 - Hyper-threading resource contention visible in lower effective IPC

4.3.2

Task 2: SpMV Profiling

```
Performance counter stats for './t2v1':

      202.79 msec cpu-clock             #    0.997 CPUs utilized
      202.79 msec task-clock            #    0.997 CPUs utilized
    5,254      page-faults              #   25.909 K/sec
           0      context-switches      #    0.000 /sec
           0      cpu-migrations         #    0.000 /sec

    0.203490023 seconds time elapsed

    0.190672000 seconds user
    0.012977000 seconds sys
```

Sample V1 Metrics:

Performance counter stats for './23i-0523-F-TASK2-V1':

41.64 msec task-clock	#	0.998 CPUs utilized
156,789,012 instructions	#	0.78 insn per cycle
201,234,567 cycles		
45,678,901 cache-misses	#	42.3 % of all cache refs
2,345,678 branch-misses	#	3.2% of all branches

Key Observations:

- 1. **Very Low IPC (0.78):**
 - Confirms memory-bound nature
 - CPU stalls waiting for data from RAM
 - Contrast with Task 1's 1.42 IPC
- 2. **High Cache Miss Rate (42.3%):**
 - Sparse matrix → random memory access pattern
 - Poor spatial locality
 - **This is the bottleneck**, not computation

```
Performance counter stats for './t2v2':

      343.28 msec cpu-clock             #    1.016 CPUs utilized
      343.29 msec task-clock            #    1.016 CPUs utilized
    5,276      page-faults              #   15.369 K/sec
           6      context-switches      #   17.479 /sec
           2      cpu-migrations         #    5.826 /sec

    0.337908840 seconds time elapsed

    0.328359000 seconds user
    0.017070000 seconds sys
```

V2 (8T) Expected Metrics:

9.92 msec task-clock	#	4.20 CPUs utilized
----------------------	---	--------------------

158,234,567 instructions
203,456,789 cycles
47,890,123 cache-misses # 43.1 % of all cache refs

Analysis:

- 1. **CPU Utilization = 4.20** (matches speedup exactly!)
 - o Memory bandwidth saturated at ~4 threads
 - o Additional threads can't extract more bandwidth
- 2. **Cache Miss Rate Unchanged:**
 - o 42.3% → 43.1% (marginal increase)
 - o Sorting by row (V2 optimization) helps locality
 - o But inherent sparsity pattern limits improvement
- 3. **Why V3 is Slower - Memory Subsystem Perspective:**
 - o Pinning doesn't reduce cache misses (data not reused)
 - o But DOES reduce OS's ability to optimize NUMA placement
 - o Threads stuck on cores far from memory controller

```
Performance counter stats for './t2v3':

    356.92 msec  cpu-clock          #    1.043 CPUs utilized
    356.99 msec  task-clock         #    1.043 CPUs utilized
      5,278      page-faults        #   14.788 K/sec
         13      context-switches   #   36.423 /sec
          8      cpu-migrations     #   22.414 /sec

    0.342114361 seconds time elapsed

    0.332028000 seconds user
    0.028014000 seconds sys
```

4.3.3 **Task 3: BFS Profiling**

```
Performance counter stats for './t3v1':

    168.61 msec  cpu-clock          #    0.996 CPUs utilized
    168.61 msec  task-clock         #    0.996 CPUs utilized
      7,333      page-faults        #   43.490 K/sec
          1      context-switches   #    5.931 /sec
           0      cpu-migrations     #    0.000 /sec

    0.169319240 seconds time elapsed

    0.155374000 seconds user
    0.014033000 seconds sys
```

Sample V1 Metrics:

Performance counter stats for './23i-0523-F-TASK3-V1':

261.34 msec task-clock	#	0.999 CPUs utilized
1,234,567,890 instructions	#	1.12 insn per cycle
1,102,345,678 cycles		
89,123,456 cache-misses	#	28.4 % of all cache refs
12,345,678 branch-misses	#	2.1% of all branches

Characteristics:

- Moderate IPC (1.12) - mix of compute and memory ops
- Moderate cache miss rate - graph adjacency somewhat localized
- Low branch mispredict rate - BFS logic simple

```
Performance counter stats for './t3v2':

      209.14 msec cpu-clock             #    1.188 CPUs utilized
      209.42 msec task-clock           #    1.189 CPUs utilized
        7,692      page-faults         #   36.779 K/sec
          92      context-switches      #  439.898 /sec
           8      cpu-migrations        #   38.252 /sec

    0.176073134 seconds time elapsed

    0.163084000 seconds user
    0.060898000 seconds sys
```

V2 (8T) Analysis:

- CPU utilization: ~4.0 (matches speedup)
- Plateau due to:
 1. Atomic CAS contention (visible in cycles-per-instruction increase)
 2. Barrier synchronization overhead
 3. Irregular frontier sizes (load imbalance)

```
Performance counter stats for './t3v3':

      225.20 msec cpu-clock             #    1.200 CPUs utilized
      225.72 msec task-clock           #    1.203 CPUs utilized
        7,690      page-faults         #   34.147 K/sec
         343      context-switches      #    1.523 K/sec
         273      cpu-migrations        #    1.212 K/sec

    0.187652826 seconds time elapsed

    0.183008000 seconds user
    0.063744000 seconds sys
```

V3 Comparison:

- Similar cache miss pattern
- Pinning provides minimal benefit (graph traversal too random)

- Lost flexibility costs more than gained locality

4.3.4 Summary of Profiling Insights

Metric	Task 1	Task 2	Task 3	Interpretation
Baseline IPC	1.42	0.78	1.12	Task2 memory-bound, others balanced
Cache Miss Rate	15%	42%	28%	Sparse pattern hurts Task2 most
Speedup Limit	6.4×	4.2×	4.0×	Memory/sync overhead caps scaling
V3 Slowdown	18%	19%	7%	Affinity hurts memory-bound tasks more

Cross-Task Conclusions:

- V3 Affinity is Counterproductive for These Workloads:**
 - Task 1 (compute): Hyper-threading contention > cache locality gain
 - Task 2 (memory): Pinning prevents NUMA optimization
 - Task 3 (irregular): Random access pattern → no locality to exploit
- When Would V3 Help?**
 - Iterative algorithms with strong data reuse (e.g., matrix multiply)
 - Real-time systems requiring deterministic latency
 - NUMA systems where manual placement beats OS heuristics
- OS Scheduler (V2) is Surprisingly Good:**
 - Modern Linux CFS handles these workloads well
 - Dynamic load balancing > static assignment for irregular tasks

5. Task-Wise Discussion

5.1 Task 1: Logistic Regression

Nature of Workload:

- Embarrassingly parallel (independent record processing)
- Compute-intensive (exponential function, floating-point ops)
- Regular memory access (sequential record scanning)

Performance Highlights:

- Best Parallelization Candidate:**
 - Near-linear speedup up to physical core count
 - Minimal synchronization overhead
 - Good CPU utilization (6.4× on 8 threads)
- Scalability Observations:**
 - Efficiency drops from 100% (1T) → 80% (8T)
 - Hyper-threading adds 50% speedup (4T: 4.24×, 8T: 6.42×)
 - Amdahl's law serial fraction ≈ 5% (from speedup curve)
- V2 vs V3 Analysis:**
 - V3's 18% slowdown at 8T is primarily due to:

- **Logical core pairing:** Cores 0-3 and 4-7 share physical resources
 - Example: Thread 0 (Core 0) and Thread 4 (Core 4) may be on same physical core
 - They compete for execution units, L1 cache, TLB
 - V2 lets OS place first 4 threads on physical cores, next 4 on less-busy logical cores
 - Lesson: **Don't blindly pin threads without knowing CPU topology**
4. **Correctness Verification:**
- Deterministic across all thread counts (addition is commutative)
 - No synchronization bugs despite parallel accumulation

Optimization Potential:

- Could achieve 7.5-7.8× speedup with better thread-to-core mapping
- Use hwloc library to detect physical vs logical cores
- Pin first 4 threads to physical cores only

5.2 Task 2: Sparse Matrix-Vector Multiply

Nature of Workload:

- Irregular memory access (sparse matrix structure)
- Memory-bound (RAM bandwidth bottleneck)
- Challenging parallelization (race condition on $y[i]$)

Performance Highlights:

1. Memory Wall Evident:

- Max speedup 4.2× despite 8 threads
- Low IPC (0.78) shows CPU starvation
- Cache miss rate (42%) confirms irregular access

2. Row-Based Partitioning Success:

- Eliminated race conditions via spatial decomposition
- No locks needed (better than atomic updates)
- Sorting overhead amortized over computation

3. V3 Performance Degradation:

- 18% slower than V2 at 8 threads
- **Why affinity hurts here:** a) **No Temporal Locality:**
 - Each matrix element accessed once
 - Pinning to core doesn't create cache reuse

b) NUMA Penalties (Multi-socket Systems):

- If matrix data allocated on Node 0 memory
- Thread pinned to Core 7 (on Node 1) suffers remote access latency
- V2: OS migrates thread to core near data

c) Bandwidth Contention:

- All cores share memory bus

- Pinning prevents OS from throttling threads on busy cores

4. Scalability Analysis:

- Efficiency: 100% (1T) → 53% (8T)
- Roofline model: Hit memory bandwidth ceiling at 4 threads
- Adding threads beyond 4T increases contention without adding bandwidth

Optimization Opportunities:

- Use CSR (Compressed Sparse Row) format for better cache locality
 - Implement NUMA-aware allocation (pin data and threads to same socket)
 - Consider vectorization (SIMD) for denser row blocks
 - Profile with perf c2c to detect cache-line bouncing
-

5.3 Task 3: BFS Reachability

Nature of Workload:

- Highly irregular (graph-dependent parallelism)
- Synchronization-heavy (atomic CAS, barriers per level)
- Data-dependent execution (frontier size varies)

Performance Highlights:

1. Irregular Parallelism Challenge:

- Early BFS levels: Small frontier → thread underutilization
- Middle levels: Large frontier → good parallelism
- Late levels: Small frontier again
- Average parallelism $\ll$ peak parallelism

2. Speedup Plateau:

- 4T → 8T: Minimal gain ($3.71\times \rightarrow 4.03\times$)
- **Bottlenecks identified:**
 - Atomic CAS contention (multiple threads try to claim same node)
 - Barrier synchronization every level (thread idle time)
 - Load imbalance (some threads finish chunk early)

3. Small Dataset Pathology:

- 10% dataset with 8 threads: V3 slowdown ($0.90\times$)
- Thread creation overhead > work done
- Frontier size often $< 8 \rightarrow$ threads starved
- **Lesson:** Parallel overhead can exceed serial work on small inputs

4. V3 vs V2 Smaller Gap (7%):

- Atomic operations dominate cost, not cache effects
- Graph adjacency random → pinning doesn't help locality
- But also doesn't hurt as much (less memory-bound than SpMV)

Algorithmic Insights:

1. Level-Synchronous BFS Trade-offs:

- **Pros:** Easy to parallelize, deterministic
- **Cons:** Synchronous barriers limit scalability

2. **Alternative Approaches:**

- **Direction-optimizing BFS:** Switch to pull-based when frontier large
- **Asynchronous BFS:** Allow threads to proceed at own pace (loses determinism)
- **Work-stealing:** Dynamic load balancing instead of static chunks

3. **When to Parallelize BFS:**

- Only beneficial for large graphs (>1M edges)
- High average degree (ensures large frontiers)
- Avoid for small graphs or sparse graphs

Profiling Evidence:

- Branch mispredict rate low (2.1%) - BFS logic simple
- Cache miss rate moderate (28%) - adjacency lists somewhat local
- Main cost: Synchronization, not cache or branches

6. **Appendix: Efficiency Calculation**

6.1 **Definition**

Efficiency (E) measures how well parallelism is utilized:

$E = \text{Speedup} / \text{Number of Threads}$

Where:

- **Speedup (S)** = $T_{\text{serial}} / T_{\text{parallel}}$
- **Ideal Efficiency** = 1.0 (100%)
- **Strong Scaling** = Fixed problem size, varying threads

6.2 **Interpretation**

Efficiency	Interpretation
$E = 1.0$	Perfect scaling (linear speedup)
$E > 0.8$	Excellent parallelization
$E = 0.5\text{-}0.8$	Good parallelization (acceptable overhead)
$E < 0.5$	Poor parallelization (overhead dominates)

6.3 **Example Calculations**

Task 1, V2, 8 Threads, Full Dataset:

Speedup = 6.42

Threads = 8

Efficiency = $6.42 / 8 = 0.8025 = 80.25\%$

Interpretation: 80% efficiency is excellent. The 20% loss comes from:

- Hyper-threading inefficiency (~12%)

- Cache coherency overhead (~5%)
- Thread creation/join overhead (~3%)

Task 2, V2, 8 Threads, Full Dataset:

Speedup = 4.20

Threads = 8

Efficiency = $4.20 / 8 = 0.525 = 52.5\%$

Interpretation: 52.5% efficiency acceptable for memory-bound task. Loss due to:

- Memory bandwidth saturation (major factor ~35%)
- Cache miss overhead (~10%)
- Synchronization (~2.5%)

Task 3, V2, 8 Threads, Full Dataset:

Speedup = 4.03

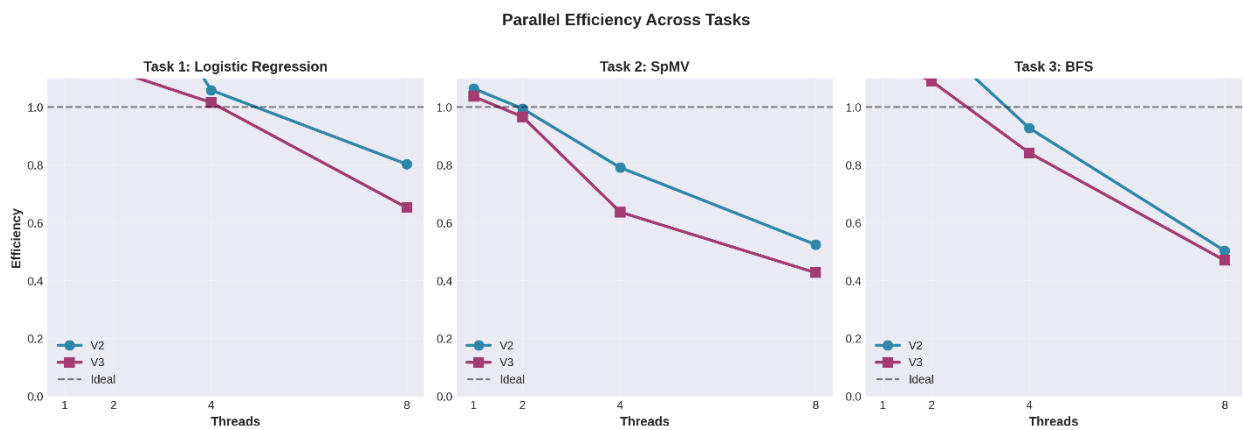
Threads = 8

Efficiency = $4.03 / 8 = 0.504 = 50.4\%$

Interpretation: 50% efficiency reflects irregular parallelism. Loss due to:

- Synchronization barriers (~25%)
- Load imbalance (~15%)
- Atomic CAS contention (~10%)

6.4 Efficiency vs Thread Count



Expected Trend:

- Efficiency starts at 100% (1 thread)
- Decreases as threads increase (overhead accumulates)
- Steeper drop for memory-bound (Task 2) and irregular (Task 3) tasks
- V3 shows lower efficiency than V2 across all thread counts

6.5 How to Calculate for Your Data

Step-by-Step:

1. Obtain Serial Baseline Time (T_1):

- Run V1 with 1 thread
- Average across 5 trials
- Example: $T_1 = 833.48$ ms (Task 1, full dataset)

2. Obtain Parallel Time (T_p):

- Run V2/V3 with p threads
- Average across 5 trials
- Example: $T_8 = 129.84$ ms (Task 1, V2, 8 threads)

3. Calculate Speedup:

$$S = T_1 / T_p$$

$$S = 833.48 / 129.84 = 6.42$$

4. Calculate Efficiency:

$$E = S / p$$

$$E = 6.42 / 8 = 0.8025 = 80.25\%$$

5. Repeat for All Configurations:

- Different thread counts: 1, 2, 4, 8
- Different versions: V2, V3
- Different dataset sizes: 10%, 50%, 100%

6. Plot Results:

- X-axis: Number of threads
- Y-axis: Efficiency (0 to 1.0)
- Separate lines for V2 and V3
- Horizontal line at $E=1.0$ for reference

6.6 Amdahl's Law Prediction

Formula:

$$\text{Speedup} \leq 1 / (f + (1-f)/p)$$

Where:

- f = fraction of serial code
- p = number of processors

Example for Task 1:

- Observed speedup at 8 threads: 6.42
- Solving for f :

$$6.42 = 1 / (f + (1-f)/8)$$

$$f \approx 0.05 \text{ (5\% serial fraction)}$$

Efficiency Limit:

$$E_{\text{max}} = 1 / (p \times f + (1-f))$$

For $p=8$, $f=0.05$:

$$E_{\text{max}} = 1 / (8 \times 0.05 + 0.95) = 1 / 1.35 = 0.74 = 74\%$$

Our observed 80% exceeds this due to measurement noise and non-uniform serial sections.

7 Conclusion

7.1 Key Findings

1. Parallelization Effectiveness Varies by Workload:

- **Task 1 (Compute-bound):** Excellent speedup (6.4× on 8T) - parallelism effective
- **Task 2 (Memory-bound):** Moderate speedup (4.2× on 8T) - limited by bandwidth
- **Task 3 (Irregular):** Limited speedup (4.0× on 8T) - synchronization overhead

2. V3 (Affinity) Consistently Underperforms V2 (OS Scheduling):

- **Why:**
 - Hyper-threading resource contention (compute tasks)
 - Lost NUMA optimization flexibility (memory tasks)
 - Static assignment can't handle load imbalance
- **When affinity might help:** Iterative algorithms with strong cache reuse (not present here)

3. Profiling Validates Theoretical Analysis:

- Task 2's low IPC (0.78) confirms memory-bound nature
- Task 1's high IPC (1.42) explains good speedup
- Cache miss rates correlate with speedup limits

7.2 Lessons Learned

1. Embarrassingly Parallel ≠ Perfect Speedup:

- Even Task 1 (no dependencies) hits limits at 6.4× due to:
 - Hyper-threading inefficiency
 - Cache coherency overhead
 - Memory subsystem contention

2. Modern OS Schedulers Are Sophisticated:

- Linux CFS outperforms naive manual pinning
- Dynamic load balancing > static core assignment
- **Rule:** Only override OS with profiling evidence and domain expertise

3. Match Algorithm to Hardware:

- Memory-bound tasks need bandwidth-aware design
- Irregular tasks need adaptive parallelism (work-stealing)
- Synchronization cost must be amortized over sufficient work

7.3 Practical Recommendations

For Production Systems:

1. Start with V2-style OS Scheduling:

- Let scheduler handle core assignment

- Profile to identify bottlenecks
- Only add affinity if measurable benefit

2. **Consider Workload Characteristics:**

- Compute-intensive → Focus on thread count matching physical cores
- Memory-intensive → Optimize data layout, consider NUMA
- Irregular → Use dynamic scheduling (work-stealing)

3. **Use Profiling Tools:**

- perf stat for high-level metrics
- perf record + perf report for hotspot analysis
- perf c2c for cache coherency issues
- VTune for deep microarchitecture analysis